

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	D.Raghava
Register Number	192525304

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I D.Raghava(192525304) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.Raghava

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

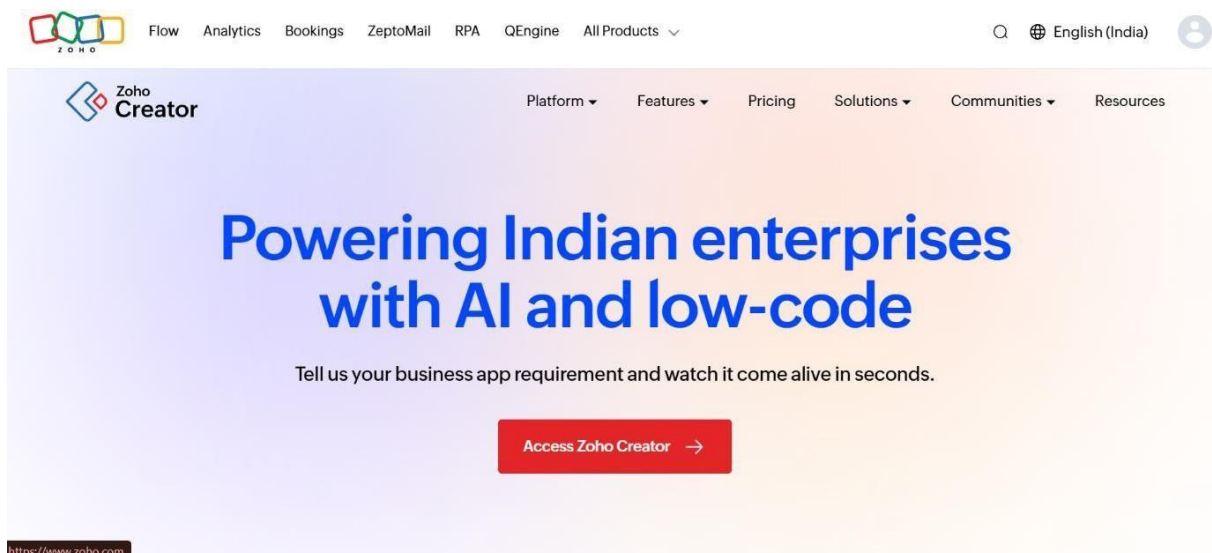
- 1) Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM:

To create a simple cloud software application for Library Book Reservation System for SIMATS Library using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

STEP 1: Log into zoho.com.



STEP 2: Create a new application from scratch in Zoho Creator.



Create from scratch

Application Name *

☐ Enable Environment [Learn More](#)

Create Application

STEP 3: Create a new application and create forms from scratch.

SIMATS Library Book Rese...

+

DesignWorkflowSettingsUpgradeAccess this application

SIMATS Library Book Re...

Start building your application by creating a form

Create New Form

Forms

ChatsContactsHere is your Smart Chat (Ctrl+Space)

Trial expires in 14 days | Upgrade | Get started faster! | Help

SIMATS Library Book Rese...

+

DesignWorkflowSettingsUpgradeAccess this application

SIMATS Library Book Re...

Start building your application by creating a form

Create New Form

Forms

ChatsContactsHere is your Smart Chat (Ctrl+Space)

Trial expires in 14 days | Upgrade | Get started faster! | Help

STEP 4: Create Book Details Form.

The screenshot shows the 'Book Details' form creation interface. The 'Basic Fields' panel on the left contains icons for Name, Email, Address, Phone, Single Line, Multi Line, Number, and Date. The main form area lists fields: Book ID (selected), Book Title, Author, Publication, Year Of Publication (highlighted in blue), Edition, ISBN Number, and Book Category. The 'Field Properties' panel on the right shows settings for the selected 'Book ID' field: Field name is 'Book ID', Field link name is 'book_id', Validation is set to 'Start index' with a value of '1', and Field type is 'Auto Number'.

The screenshot shows the 'Book Reservation' form creation interface. The 'Basic Fields' panel on the left contains icons for Name, Email, Address, Phone, Single Line, Multi Line, Number, and Date. The main form area lists fields: Student (selected), Book, Reservation Date, Expected Issue Date, Reservation Status, and Rack Number. The 'Field Properties' panel on the right shows settings for the selected 'Student' field: Field name is 'Student', Field link name is 'student', and Display type is 'Multi Select'.

The screenshot shows the 'Student Details' form creation interface. The 'Basic Fields' panel on the left contains icons for Name, Email, Address, Phone, Single Line, Multi Line, Number, and Date. The main form area lists fields: Student ID (selected), Student Name, Registration Number, Department, Year Of Study, Mobile Number, Email, and Status. The 'Field Properties' panel on the right shows settings for the selected 'Student ID' field: Field name is 'Student ID', Field link name is 'student_id', Validation is set to 'Start index' with a value of '1', and Field type is 'Auto Number'.

STEP 6: Create Book Reservation Form

SIMATS Library Book Reservation – Book Issue Return

+

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123
Number

15
Date

Student

🗑️

Book

📅

Issue Date

📅

Due Date

📅

Return Date

🏷️

Status

🔍

Field Properties

Field name

Student

Field link name

student

[View Field References](#)

Display Fields

SIMATS Library Book Reservation System → Student Details

Student ID

▼

+

–

+

Display type

Multi Select

SIMATS Library Book Reservation...
Book Search

+ Done

Basic Fields

Name	Email
Address	Phone
Single Line	Multi Line
Number	Date

Book Title

Author

Book Category

Publication

Availability

Field Properties

Field name

Field link name

[View Field References](#)

Validation
☐ Mandatory ☐ No duplicate values

Initial value

Character maximum

SIMATS Library Book Reservation...
Feedback

+

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number

Date

Student

Book

Rating

Comments

Field Properties

Field name

Student

Field link name

student

[View Field References](#)

Display Fields

SIMATS Library Book Reservation System → Student Details

Student ID

+

-

↕

Display type

Multi Select

STEP 9: Create Feedback form

STEP 10: Add sample data to all the created forms.

SIMATS Library Book Reservation System

Trial expires in 14 days Upgrade Edit this application Help

SIMATS Library Book Reser...

Dashboard

Book Reservation

Book Issue Return

Feedback

Student Details

+ Student Details

Student Details Re...

Board

Book Details

Kanmani Log...

Student Details

Student Name Kanmani
First Name

Registration Number 192511414

Department Computer Science

Year Of Study II Year

Mobile Number +91 8765432199

Email kanmanilogu04@gmail.com

Status ☒ Active ☐ Inactive ☐ Other

SIMATS Library Book Reservation System

Trial expires in 14 days Upgrade Edit this application Help

SIMATS Library Book Reser...

Dashboard

Book Reservation

Book Issue Return

Feedback

Student Details

+ Student Details

Student Details Re...

Board

Book Details

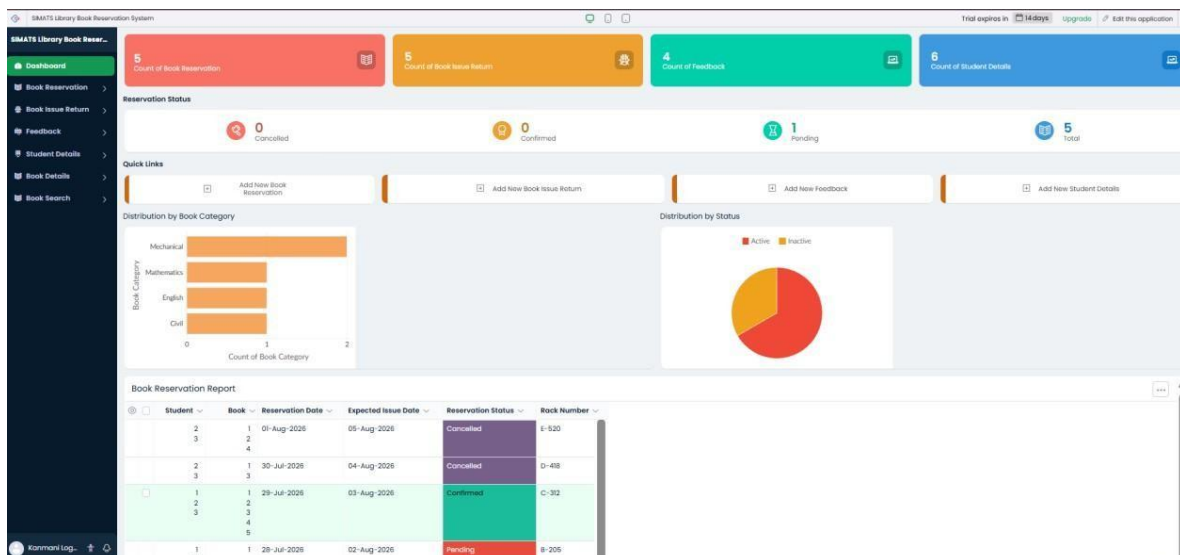
Kanmani Log...

Student Details Report

Student ID	Student Name	Registration Number	Department	Year Of Study	Mobile Number	En
6	Kanmani	192511414	Computer Science	II Year	+918776543278	kc
5	Eva Garcia	STU2026005	Civil	III Year	+17179895133	ev
4	David Lee	STU2026004	Electronics	II Year	+17059044102	dc
3	Carol Martinez	STU2026003	Mathematics	IV Year	+18887424338	cc
2	Bob Williams	STU2026002	Mechanical	III Year	+19179771587	bc
1	Alice Johnson	STU2026001	Artificial Intelligence	III Year	+18202784391	al

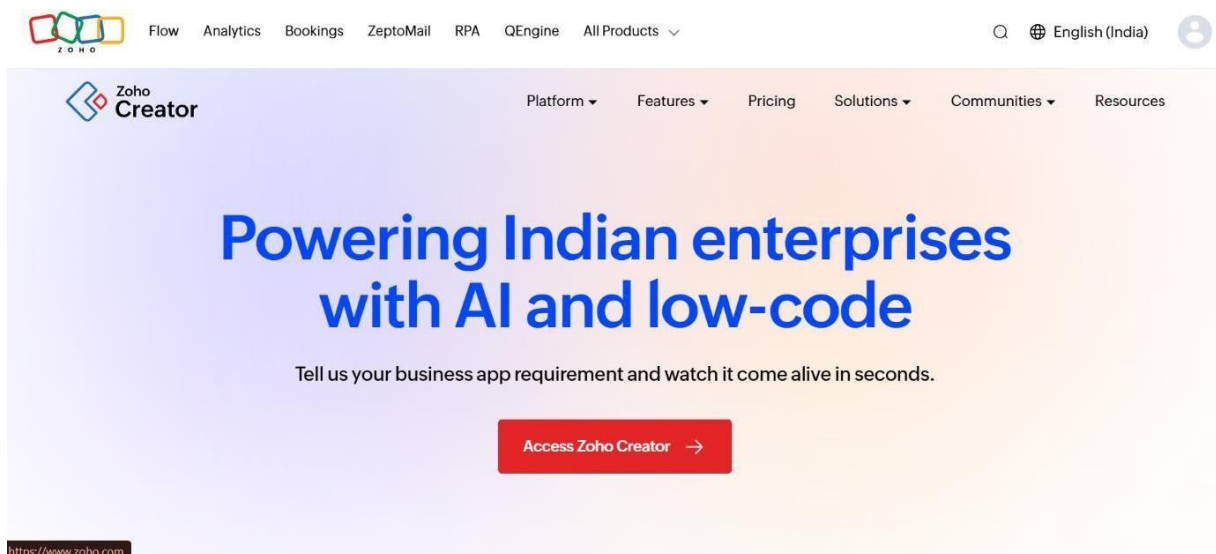
Showing 6 of 6

STEP 12: Create and display the Dashboard with important library information and book reservation details.



2) Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

STEP 1: Log into zoho.com.



STEP 2: Create a new application from scratch in Zoho Creator.



Create from scratch

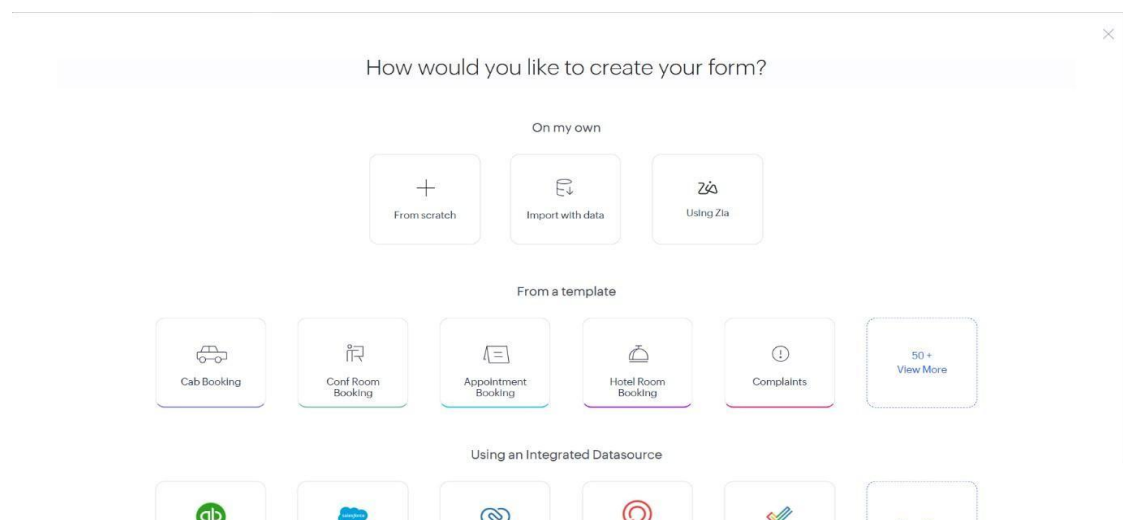
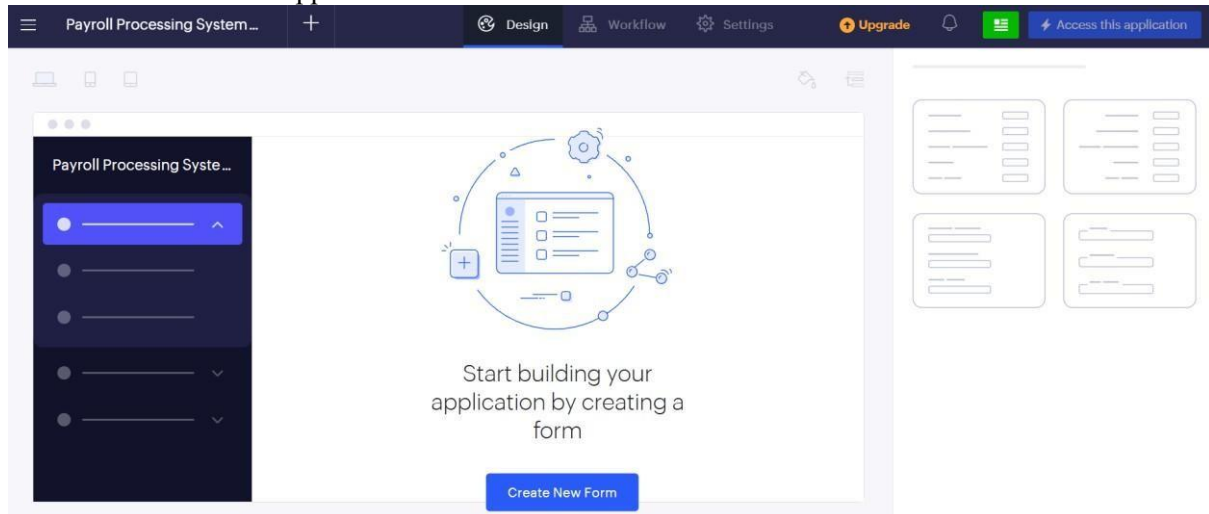
Application Name *

Payroll Processing System

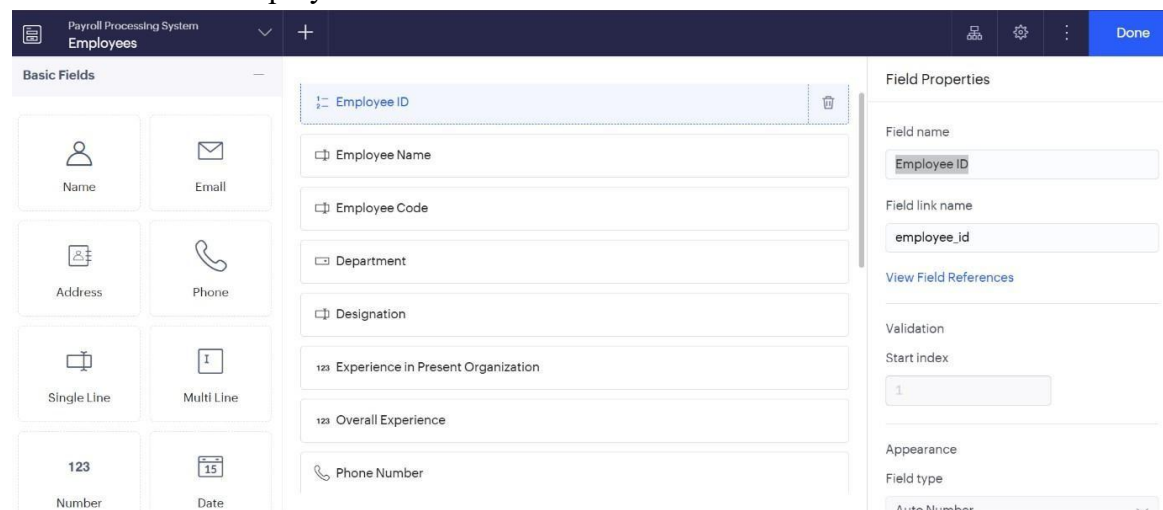
☐ Enable Environment [Learn More](#)

Create Application

STEP 3: Create a new application and create forms from scratch.



STEP 4: Create Employee Form.



STEP 5: Create Payroll records Form.

Payroll Processing System

Payroll Records

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Payroll ID

Employee

Salary Month

Basic Salary

Bonus

Deductions

Gross Salary

Net Salary

Field Properties

Field name

Payroll ID

Field link name

payroll_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

Here is your Smart Chat (Ctrl+Space)

Trial expires in 14 days | Upgrade | Get started faster! | Help

STEP 6: Create Payroll Records form

Payroll Processing System

Payment Records

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Employee

Payroll

Payment Date

Payment Method

Payment Status

Field Properties

Field name

Employee

Field link name

employee

View Field References

Display Fields

Payroll Processing System → Employees

Employee ID

Display type

Multi Select

STEP 7: Create Leave records form.

Payroll Processing System

Leave Records

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Employee

Leave Type

From Date

To Date

Leave Status

Field Properties

Field name

Employee

Field link name

employee

View Field References

Display Fields

Payroll Processing System → Employees

Employee ID

Display type

Multi Select

STEP 8: Add sample data to all the created forms.

Payroll Processing System

Trial expires in 14 days Upgrade Edit this application Help

Payroll Processing System

- Dashboard
- Payment Records
- Leave Records
- + Leave Records**
- Leave Records Rep...
- Board
- Payroll Records
- Employees

Leave Records

Employee: 4 x

Leave Type: Sick Leave x v

From Date: 28-Jul-2026

To Date: 29-Jul-2026

Leave Status: Approved x v

Submit Reset

Payroll Processing System

Trial expires in 14 days Upgrade Edit this application Help

Payroll Processing System

- Dashboard
- Payment Records
- + Payment Records**
- Payment Records ...
- Board
- Leave Records
- Payroll Records
- Employees

Payment Records

Employee: 3 x

Payroll: 4 x

Payment Date: 28-Jul-2026

Payment Method: Cash x v

Payment Status: ☐ Paid ☒ Pending ☐ Other

Submit Reset

Kanmani Log...

STEP 11: Create reports to display the data entered in the respective forms.

Payroll Processing System

Trial expires in 14 days Upgrade Edit this application Help

Payroll Processing System

- Dashboard
- Payment Records
- + Payment Records**
- Payment Records ...
- Board
- Leave Records
- Payroll Records
- Employees

Payment Records Report

	Employee	Payroll	Payment Date	Payment Method	Payment Status
	3	3			
		4			
		5			
	1	1	28-Jul-2026	Bank Transfer	Paid
	2	2			
		3			
		4			
	2	2	28-Jul-2026	UP	Paid
		3			
		4			
...	1	1	28-Jul-2026	Cash	Paid
	2	3			
	3				
	4				
	3	4	28-Jul-2026	Cash	Pending

Showing 6 of 6

Kanmani Log...

Payroll Processing System

Trial expires in 14 days Upgrade Edit this application Help

Payroll Processing System

Dashboard

Payment Records

Leave Records

Leave Records

Leave Records Rep...

Board

Payroll Records

Employees

Kanmani Log...

Leave Records Report

Employee

Leave Type

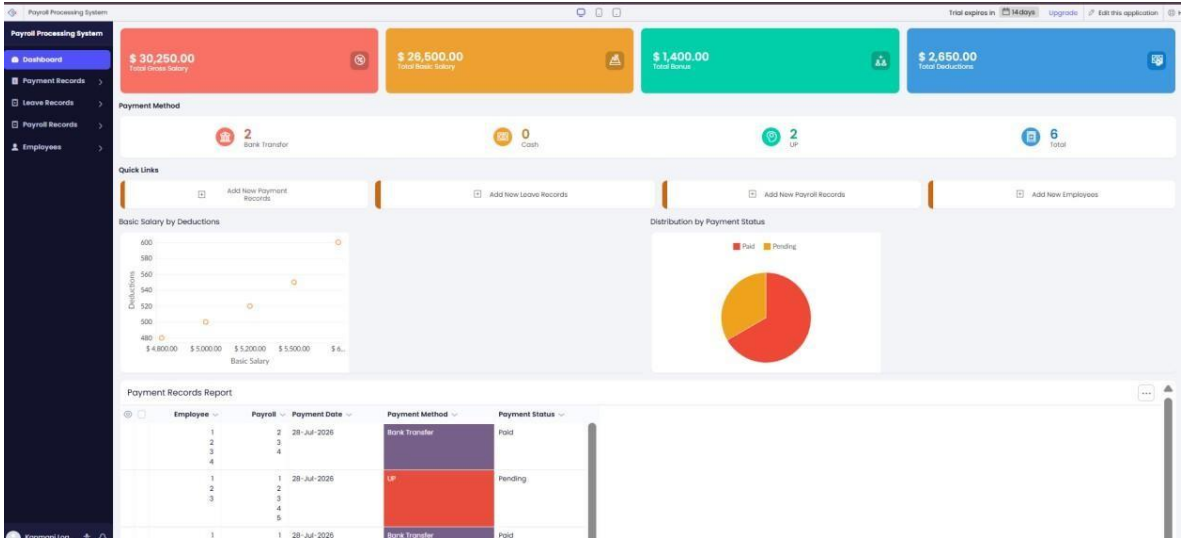
From Date

To Date

Leave Status

4	Sick Leave	28-Jul-2026	29-Jul-2026	Approved
1	Casual Leave	25-Jul-2026	28-Jul-2026	Rejected
2				
3				
1	Emergency Leave	20-Jul-2026	22-Jul-2026	Approved
2				
3				
1	Sick Leave	10-Jul-2026	12-Jul-2026	Pending
1	Casual Leave	01-Jul-2026	05-Jul-2026	Approved
2				

Showing 5 of 5



STEP 12: Create and display the Dashboard

3) Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux).

PROCEDURE:

STEP 1:Download VMware workstation and installed as type 2hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

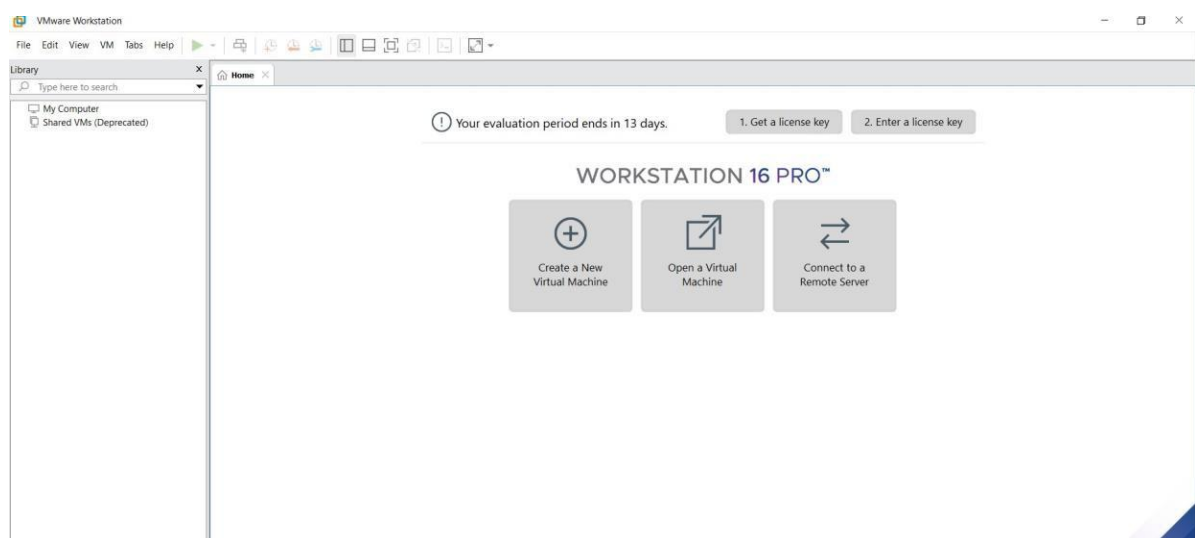
STEP 3: In VMware workstation->create new VM.

STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine. **STEP 6:** Launch the VM.

IMPLEMENTATION:

STEP 1:DOWLOAD VMWARE WORKSTATION AND INSTALLED AS TYPE 2HYPERVISOR



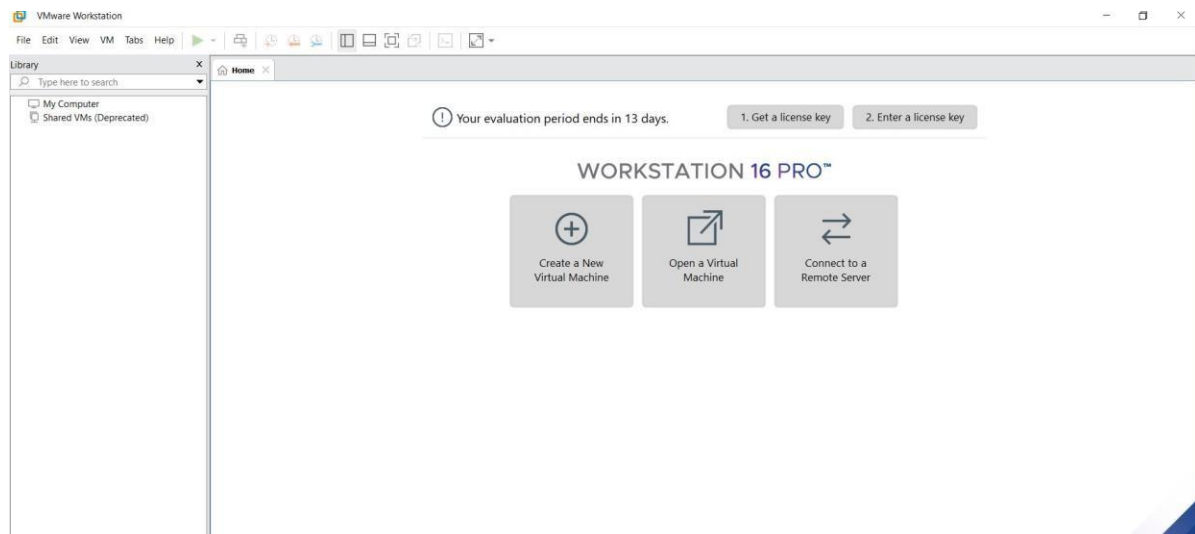
STEP2: DOWNLOAD LIBUNTU OR TINY OS AS ISO IMAGE FILE

STEP8: Add sample data to all the created forms.

Index of /11.x/x86/release/

distribution_files/	09-Feb-2020 11:50	-
src/	03-Dec-2019 11:14	-
Core-11.1.iso	01-Apr-2020 07:49	14757888
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48
Core-11.1.iso.zsync	01-Apr-2020 07:49	50639
Core-current.iso	01-Apr-2020 07:49	14757888
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	369358
CorePlus-current.iso	01-Apr-2020 07:50	216006656
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	68301
TinyCore-current.iso	01-Apr-2020 07:50	19922944

STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM



STEP 4: DO THE BASIC CONFIGURATION SETTINGS.



STEP 8: Add sample data to all the created forms.



Your evaluation period ends in 12 days.

1. Get a license key

2. Enter a license key

New Virtual Machine Wizard x 6 PRO™

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:
No drives available

☐ Installer disc image file (iso):
C:\Users\Gopi Prasad-PC\Downloads\ubuntu-20.04.2... Browse...

☒ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel

Connect to a Remote Server

New Virtual Machine Wizard x 6 PRO™

Select a Guest Operating System
Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows

☒ Linux

☐ VMware ESX

☐ Other

Version

Ubuntu 64-bit

Help < Back Next > Cancel

Connect to a Remote Server

New Virtual Machine Wizard x 6 PRO™

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:
Ubuntu Vmware

Location:
C:\Users\Gopi Prasad-PC\Documents\Virtual Machines\Ubuntu \ Browse...

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

Connect to a Remote Server

STI

New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

Location:

The default location can be changed at Edit > Preferences.

< Back Next > Cancel



! Your evaluation period ends in 12 days.

1. Get a license key

2. Enter a license key

! Your evaluation period ends in 12 days.

1. Get a license key

2. Enter a license key

New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file
☐ Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel



New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

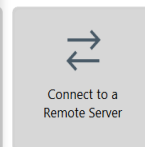
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☐ Split virtual disk into multiple files
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Help < Back Next > Cancel



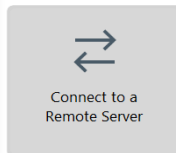
New Virtual Machine Wizard

Ready to Create Virtual Machine
Click Finish to create the virtual machine. Then you can install Ubuntu 64-bit.

The virtual machine will be created with the following settings:

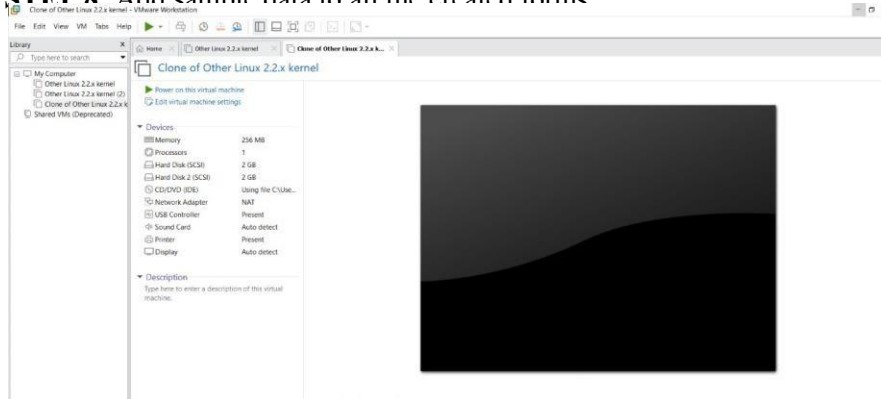
Name:	Ubuntu Vmware
Location:	C:\Users\Gopi Prasad-PC\Documents\Virtual Machines\U...
Version:	Workstation 16.x
Operating System:	Ubuntu 64-bit
Hard Disk:	20 GB
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	2 CPU cores, CD/DVD, USB Controller, Printer, Sound C...

< Back Finish Cancel



STEP 5: CREATED TINYOS VIRTUAL MACHINE

STEP 8: Add sample data to all the created forms



4) virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Sol) AIM:

To create a virtual hard disk and allocate the storage using vm ware workstation

PROCEDURE:

STEP 1:GOTO VM WARE WORKSTATION.

STEP2: RIGHT CLICK THE VM AND GOTO THE SETTINGS.

STEP 3: ADD HARDWARE WIZARD AND SELECT SCSI AND CLICK NEXT.

STEP 4: CREATE NEW VIRTUAL DISK.

STEP 5: SELCT THE DISK SIZE AS 2.0. AND SELCT SPLIT VIRTUAL DISK INTOMULTIFILES.

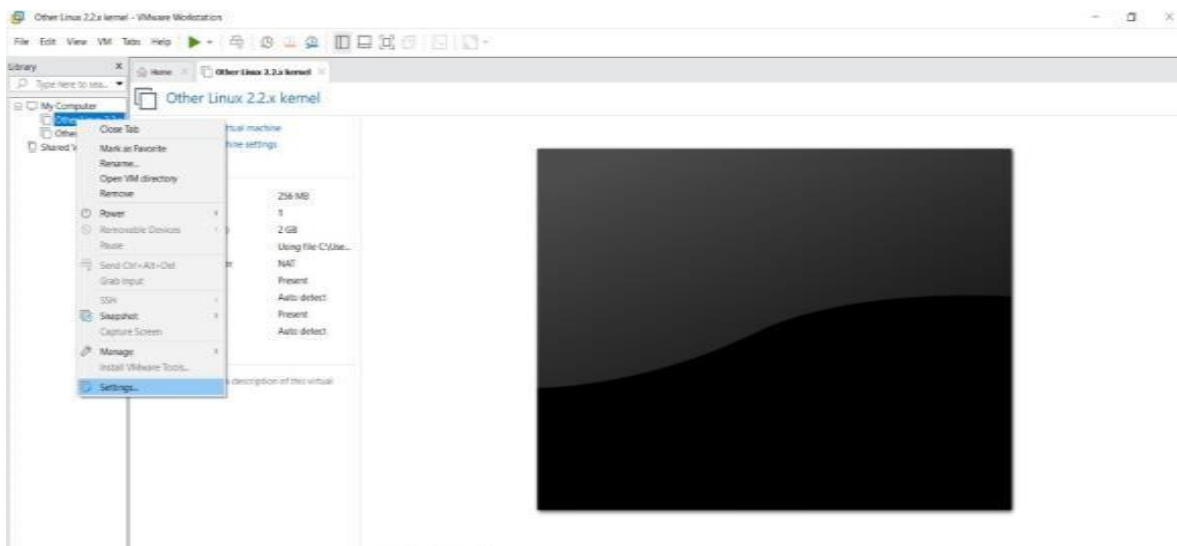
STEP 6: GIVE NAME AND CLICK THE FINISH.

STEP 8: Add sample data to all the created forms.

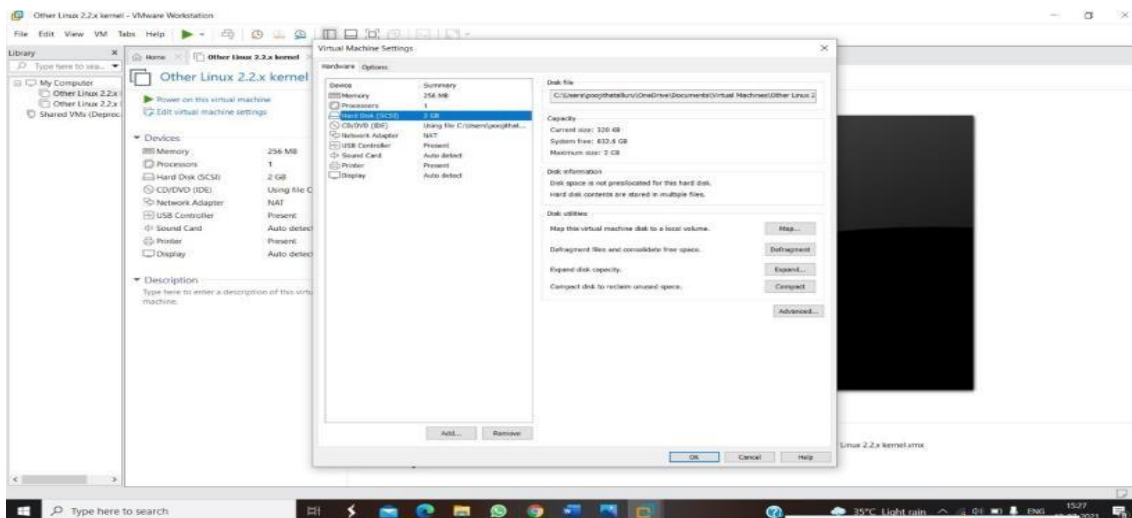
IMPLEMENTATION:

STEP 1:GOTO VM WARE WORKSTATION

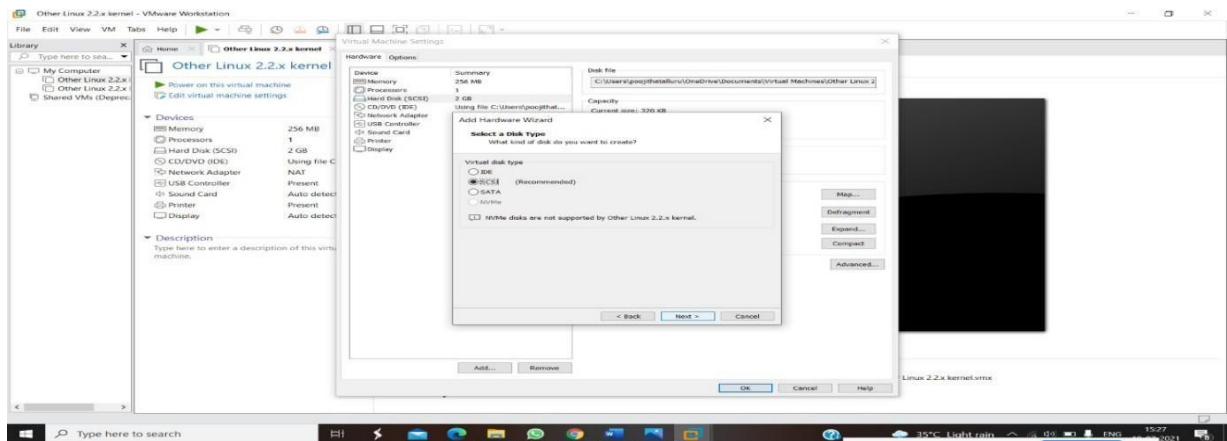
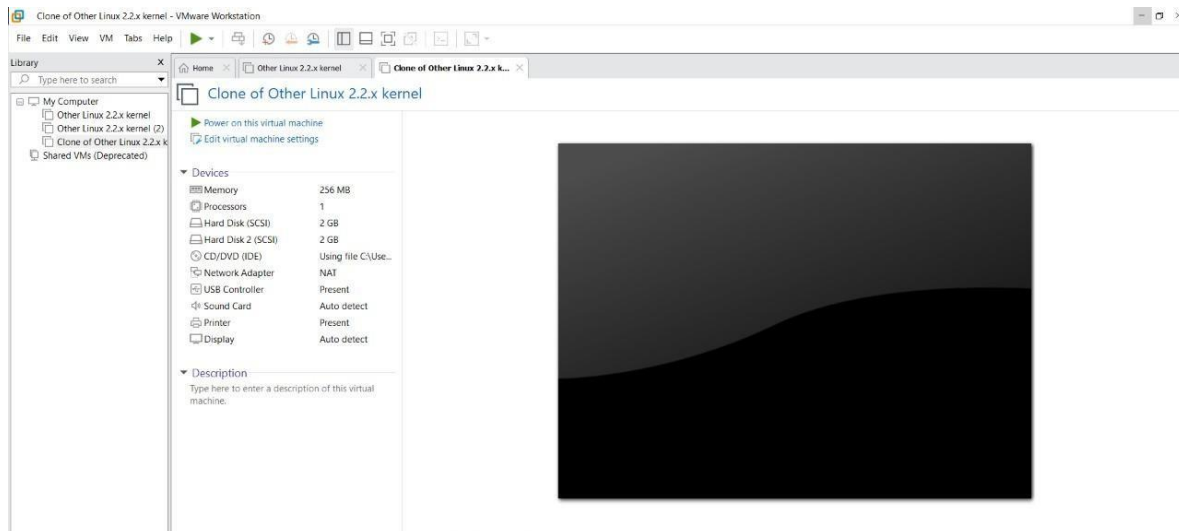
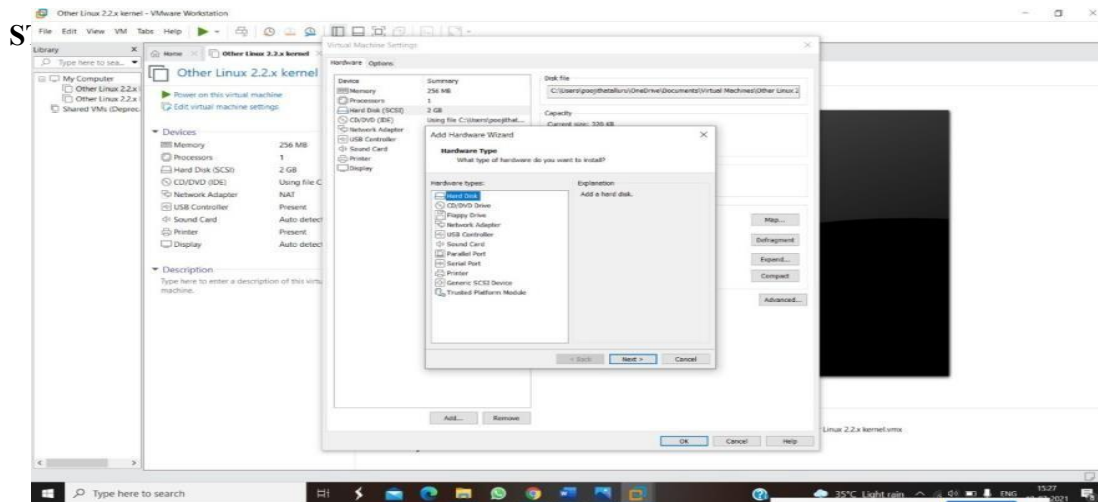
STEP2: RIGHT CLICK THE VM AND GOTO THE SETTINGS



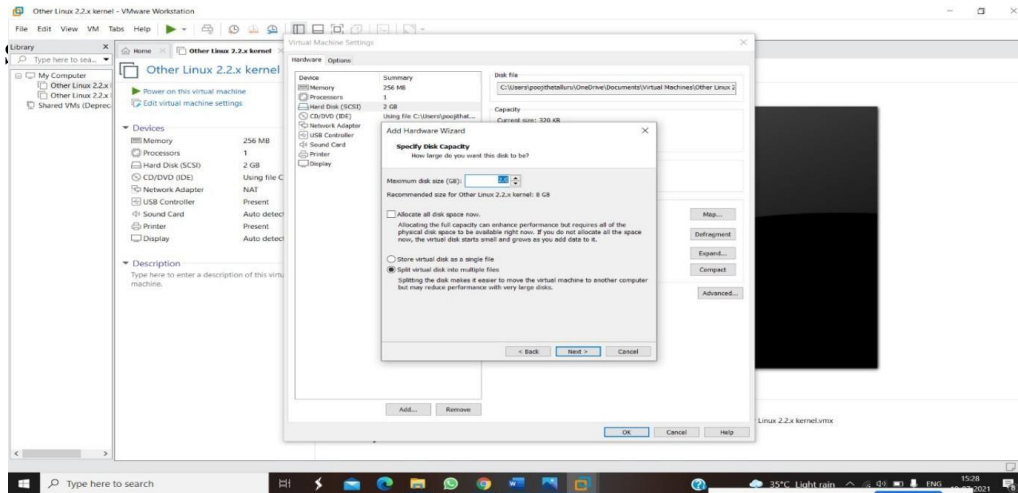
STEP 3: ADD HARDWARE WIZARD AND SELECT SCSI AND CLICK NEXT



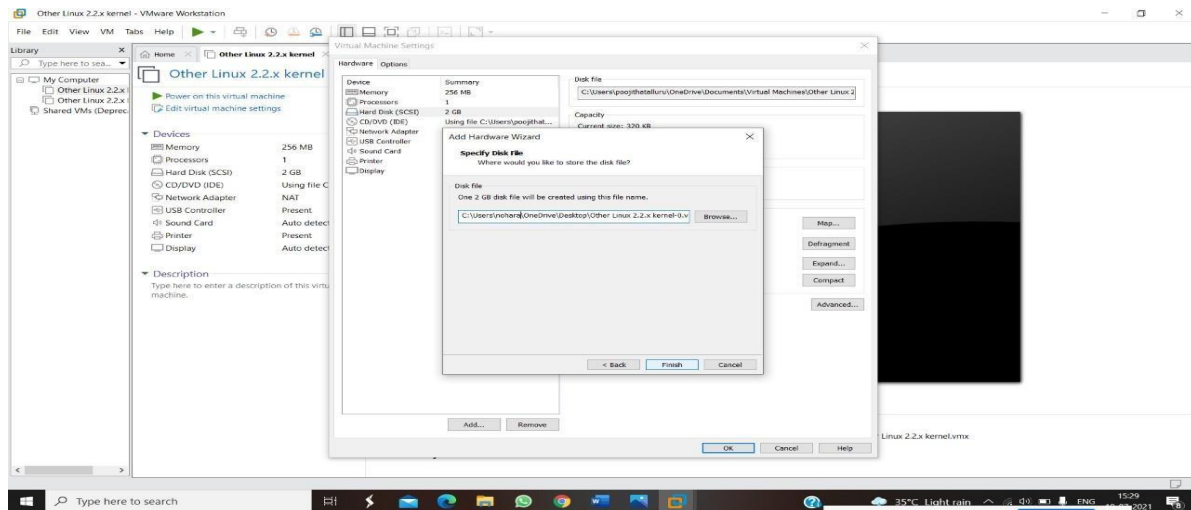
STEP 4: CREATE NEW VIRTUAL DISK



STEP 5: SELECT THE DISK SIZE AS 2.0. AND SELECT SPLIT VIRTUAL DISK INTO MULTIFILES.



STEP 6: GIVE NAME AND CLICK THE FINISH



STEP 8: Add sample data to all the created forms.